ITM SKILLS UNIVERSITY

School of Future Tech

Case Study Report

On

Currency Exchange & Forex Simulation System

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1. Introduction to the Case Study

Currency exchange plays a vital role in international trade, travel, online transactions, and global finance. Foreign exchange (Forex) systems are responsible for converting one currency into another based on current exchange rates. Accuracy, reliability, and proper transaction logging are essential in such systems.

This case study focuses on designing and implementing a Currency Exchange & Forex Simulation System using C++. The system allows users to view exchange rates, update currency rates, perform currency conversions, and store transaction records. It demonstrates how financial logic can be implemented using structured programming concepts and file handling in C++.

The project highlights the practical use of data structures, validation techniques, conversion formulas, and persistent storage mechanisms in building a functional financial simulation system.

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2. Problem Statement / Case Background (Abstract)

Background

In real-world financial systems, currency exchange operations must be precise and reliable. Exchange rates fluctuate, and incorrect calculations can result in financial loss. Additionally, maintaining transaction history is essential for transparency and auditing.

Traditional manual calculations are inefficient and error-prone. Therefore, a computerized forex simulation system is required to automate currency conversion and maintain records securely.

Abstract

This case study presents the design and implementation of a Forex Simulation System using C++. The system maintains predefined currency exchange rates relative to a base currency (INR). It allows users to update rates (except the base currency), convert between any two supported currencies, and log transactions into a file.

The system ensures input validation, case-insensitive currency handling, and proper file-based transaction storage. Through this implementation, the project demonstrates the use of structured programming, file handling, time logging, and conversion algorithms to simulate real-world forex operations in a simplified manner.

3. Case Study Design

The Forex Simulation System is designed as a menu-driven console application with the following modules:

1. Currency Rate Management

- Stores 10 predefined currencies.
- Uses INR as the base currency.
- Allows updating exchange rates (except INR).
- Validates positive rate values.

2. Currency Conversion Module

- Accepts source and target currency codes.
- Converts input amount using INR as intermediate currency.
- Validates currency codes.
- Ensures the amount entered is positive.

3. Transaction Logging Module

- Saves transaction details in a text file.
- Records:
    - Date and time
    - Source currency
    - Target currency
    - Original amount

    - Converted amount

- Allows viewing complete transaction history.

4. User Interface

- Menu-driven console interface.
- Repeats until the user selects exit.
- Displays appropriate success and error messages.

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4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key Methods and Techniques Used

1. Case-Insensitive Input Handling

- transform() function from the algorithm library.
- Converts currency codes to uppercase.

2. Currency Search Algorithm

- Linear search method.
- Searches currency code inside a predefined array.

3. Conversion Algorithm

Step 1: Convert source currency to INR
amountInINR = amount × rateToINR

Step 2: Convert INR to target currency
result = amountInINR ÷ rateToINR

This ensures accurate cross-currency conversion.

4. File Handling Techniques

- ofstream used to append transaction records.
- ifstream used to read transaction history.
- Time logging using ctime() function.

Technology Stack Used for the Case

- Programming Language: C++
- Concepts Used:
    - Structures
    - Arrays
    - Functions
    - File Handling
    - Time Library
    - Input Validation
- Development Environment:
    - VS Code / CodeBlocks / Any C++ IDE
- Data Storage:
    - Text file (transactions.txt)

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5. Problem Statement / Case Study Implementation Details and Snapshots

The implementation is organized into logical functions:

1. Currency Structure

Stores currency code and rate relative to INR.

2. showRates()

Displays all current exchange rates.

3. updateRate()

Allows updating exchange rate except base currency.

4. convertCurrency()

Performs conversion and calls transaction saving function.

5. saveTransaction()

Appends transaction details to transactions.txt.

6. viewTransactions()

Reads and displays stored transaction history.

<u>Snapshots</u> :

```
===== Forex Simulation System =====
1. Show Exchange Rates
2. Update Exchange Rate
3. Convert Currency
4. View Transaction History
5. Exit
Enter your choice:
```

```
--- Current Exchange Rates (Base: INR) ---
1 USD = 83 INR
1 EUR = 90 INR
1 GBP = 105 INR
1 JPY = 0.55 INR
1 AUD = 55 INR
1 CAD = 60 INR
1 CHF = 92 INR
1 CNY = 11.5 INR
1 SGD = 61 INR
1 INR = 1 INR
```

```
Enter your choice: 2
Enter currency code to update: USD
Enter new rate (1 USD = ? INR): 91.61
Rate updated successfully!
```

```
Enter your choice: 3
Enter FROM currency code: usd
Enter TO currency code: inr
Enter amount: 4

4 USD = 366.44 INR
Transaction saved successfully!
```

```
forex.cpp
transactions.txt    M
```

```
Date: Thu Mar  5 08:42:31 2026
From: USD
To: INR
Amount: 3
Converted: 249
-------------------------
Date: Thu Mar  5 08:46:48 2026
From: AUD
To: USD
Amount: 2
Converted Amount: 1.5694
----------------------------------
Date: Thu Mar  5 09:47:22 2026
From: USD
To: INR
Amount: 4
Converted Amount: 366.44
----------------------------------
```

```
Enter your choice: 4

--- Transaction History ---
Date: Thu Mar  5 08:42:31 2026
From: USD
To: INR
Amount: 3
Converted: 249
-------------------------
Date: Thu Mar  5 08:46:48 2026
From: AUD
To: USD
Amount: 2
Converted Amount: 1.5694
----------------------------------
Date: Thu Mar  5 09:47:22 2026
From: USD
To: INR
Amount: 4
Converted Amount: 366.44
----------------------------------
```

6. Problem Statement / Case Study Results and Conclusion

Findings

- The system successfully performs cross-currency conversions.
- Input validation prevents invalid currency codes and negative amounts.

- Transaction logging ensures record persistence.
- Using INR as a base simplifies multi-currency conversion logic.
- Case-insensitive handling improves user experience.

Conclusion

This case study demonstrates the successful implementation of a Currency Exchange & Forex Simulation System using C++. The project integrates currency management, exchange calculation, and transaction logging into a structured and functional console application.

The system effectively simulates real-world forex operations in a simplified academic model and demonstrates practical application of structured programming, file handling, and financial calculation logic.

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7. References

- Official C++ Documentation
- Documentation for <iostream>, <fstream>, <ctime>, and <algorithm> libraries
- C++ Programming textbooks and academic resources
- Online tutorials on file handling and currency conversion logic